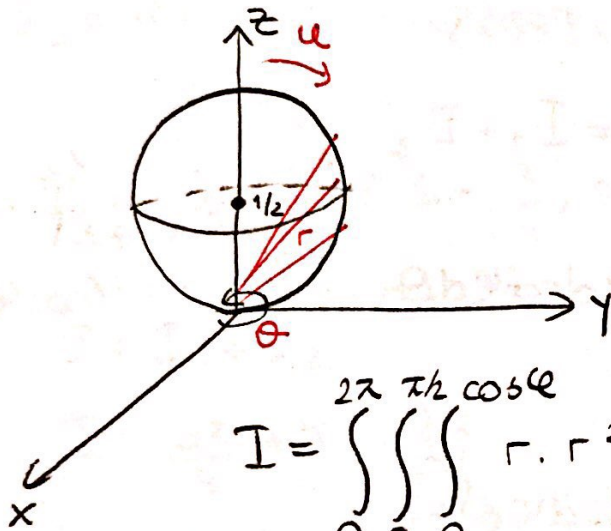


6) Evaluate $I = \iiint_V \sqrt{x^2+y^2+z^2} dv$ on the surface $x^2+y^2+z^2=z$. (Use spherical coord.)

$$\left. \begin{aligned} * x &= r \sin \varphi \cos \theta \\ y &= r \sin \varphi \sin \theta \\ z &= r \cos \varphi \end{aligned} \right\} J = r^2 \sin \varphi$$



$$\begin{aligned} * x^2 + y^2 + z^2 - z &= 0 \\ x^2 + y^2 + \left(z - \frac{1}{2}\right)^2 &= \frac{1}{4} \end{aligned}$$

$$\begin{aligned} * x^2 + y^2 + z^2 &= r^2 \\ r^2 &= z = r \cos \varphi \\ r &= \cos \varphi \quad (\text{and } r=0) \end{aligned}$$

$$I = \int_0^{2\pi} \int_0^{\pi/2} \int_0^{\cos \varphi} r \cdot r^2 \sin \varphi dr d\varphi d\theta$$

$$I = \int_0^{2\pi} \int_0^{\pi/2} \int_0^{\cos \varphi} r^3 \sin \varphi dr d\varphi d\theta = \int_0^{2\pi} \int_0^{\pi/2} \frac{r^4}{4} \sin \varphi \Big|_0^{\cos \varphi} d\varphi d\theta$$

$$I = \frac{1}{4} \int_0^{2\pi} \int_0^{\pi/2} \cos^4 \varphi \sin \varphi d\varphi d\theta, \quad \begin{aligned} \cos \varphi &= t \\ -\sin \varphi d\varphi &= dt \end{aligned}$$

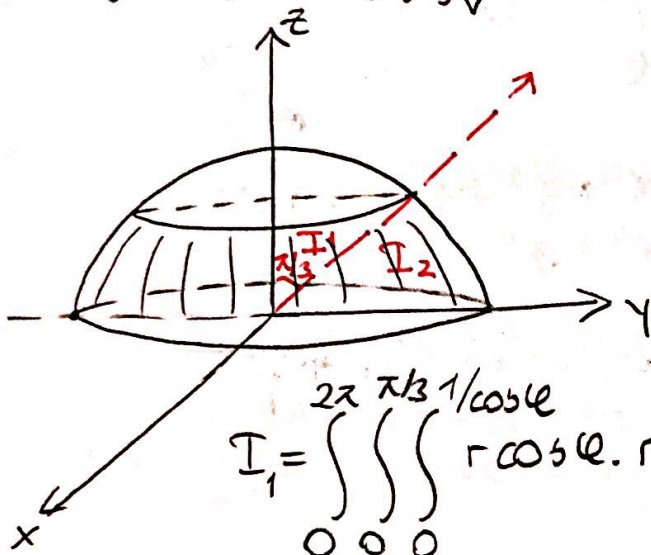
$$I = \frac{1}{4} \int_0^{2\pi} \int_1^0 t^4 (-dt) d\theta = \frac{1}{4} \int_0^{2\pi} \int_0^1 t^4 dt d\theta$$

$$I = \frac{1}{4} \int_0^{2\pi} \left. \frac{t^5}{5} \right|_0^1 d\theta = \frac{1}{20} \int_0^{2\pi} d\theta = \frac{1}{20} \cdot 2\pi$$

$$I = \frac{\pi}{10}$$

7) $V : \begin{cases} x^2 + y^2 + z^2 = 4 \\ z=1, z=0 \end{cases}$ is given. Evaluate the

integral $I = \iiint_V z \, dV$ by using spherical coordinates.



$$\begin{cases} x = r \sin \varphi \cos \theta \\ y = r \sin \varphi \sin \theta \\ z = r \cos \varphi \end{cases} \quad J = r^2 \sin \varphi$$

$$I = I_1 + I_2$$

$$I_1 = \int_0^{2\pi} \int_0^{\pi/3} \int_{1/\cos \varphi}^{2} r \cos \varphi \cdot r^2 \sin \varphi \, dr \, d\varphi \, d\theta$$

* for $z=1$
 $r = 1/\cos \varphi$

$$I_2 = \int_0^{2\pi} \int_{\pi/3}^{\pi/2} \int_0^2 r \cos \varphi \cdot r^2 \sin \varphi \, dr \, d\varphi \, d\theta$$

* for $x^2 + y^2 + z^2 = 4$
 $r^2 = 4 \Rightarrow r = 2$

$$I_1 = \int_0^{2\pi} \int_0^{\pi/3} \int_{1/\cos \varphi}^2 r^3 \sin \varphi \cos \varphi \, dr \, d\varphi \, d\theta = \frac{1}{4} \int_0^{2\pi} \int_0^{\pi/3} r^4 \sin \varphi \cos \varphi \Big|_{1/\cos \varphi}^2 d\varphi \, d\theta$$

$$I_1 = \frac{1}{4} \int_0^{2\pi} \int_0^{\pi/3} \frac{1}{\cos^4 \varphi} \cdot \sin \varphi \cos \varphi \, d\varphi \, d\theta$$

$$I_1 = \frac{1}{4} \int_0^{2\pi} \int_0^{\pi/3} \frac{\sin \varphi}{\cos^3 \varphi} \, d\varphi \, d\theta, \quad \begin{aligned} \cos \varphi &= t \\ -\sin \varphi \, d\varphi &= dt \end{aligned}$$

$$I_1 = \frac{1}{4} \int_0^{2\pi} \left(\int_1^{1/2} -\frac{dt}{t^3} \right) d\theta = \frac{1}{4} \int_0^{2\pi} \frac{1}{2} \cdot \frac{1}{t^2} \Big|_1^{1/2} d\theta$$

$$I_1 = \frac{1}{8} \int_0^{2\pi} (4 - 1) d\theta = 3 \cdot \frac{1}{8} \cdot 2\pi \Rightarrow I_1 = \frac{3\pi}{4}$$

$$I_2 = \int_0^{2\pi} \int_{\pi/3}^{\pi/2} \int_0^2 r^3 \sin \phi \cos \phi dr d\phi d\theta$$

$$I_2 = \frac{1}{4} \int_0^{2\pi} \int_{\pi/3}^{\pi/2} r^4 \cdot \sin \phi \cos \phi \Big|_0^2 d\phi d\theta = \frac{1}{4} \int_0^{2\pi} \int_{\pi/3}^{\pi/2} 16 \sin \phi \cos \phi d\phi d\theta$$

$$I_2 = 4 \int_0^{2\pi} \int_{\pi/3}^{\pi/2} \sin \phi \cos \phi d\phi d\theta = 4 \int_0^{2\pi} \frac{\sin^2 \phi}{2} \Big|_{\pi/3}^{\pi/2} d\theta$$

$$I_2 = -2 \int_0^{2\pi} \left(1 - \frac{3}{4}\right) d\theta = 2 \cdot \frac{1}{4} \cdot 2\pi \Rightarrow \underline{I_2 = \pi}$$

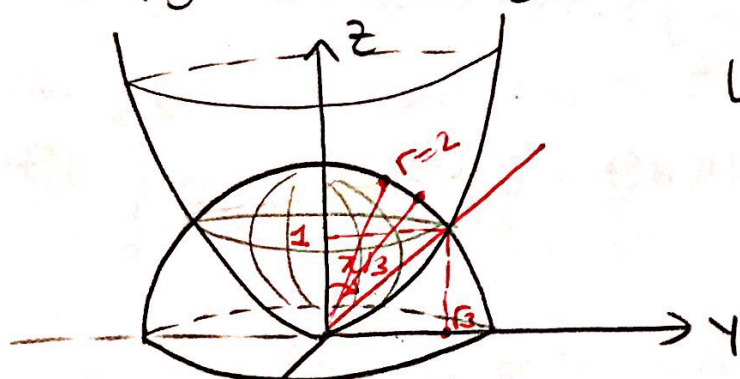
$$I = I_1 + I_2$$

$$I = \frac{3\pi}{4} + \pi = \underline{\underline{\frac{7\pi}{4}}}$$

8) Evaluate $I = \iiint_V \frac{dv}{x^2 + y^2 + z^2}$ on $V: \begin{cases} z = \sqrt{4 - x^2 - y^2} \\ x^2 + y^2 - 3z = 0 \end{cases}$

$$x^2 + y^2 + z^2 = 4 \text{ but } z \geq 0$$

$$x^2 + y^2 - 3z = 0 \Rightarrow 3z = x^2 + y^2$$



let use spherical coordinates;

$$* x^2 + y^2 + z^2 = r^2$$

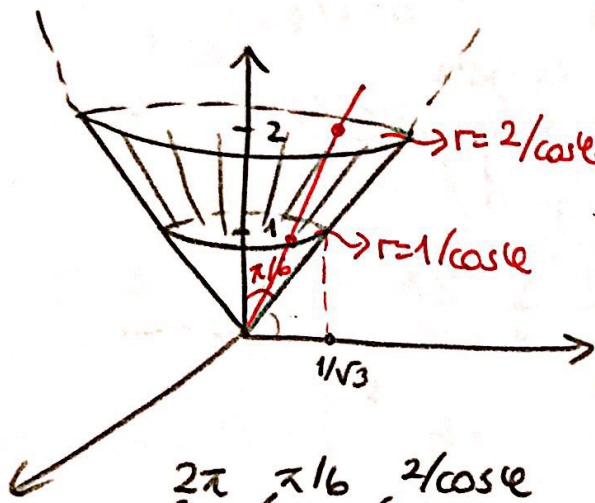
$$* 3z = x^2 + y^2 \Rightarrow r = \frac{3 \cos \phi}{\sin^2 \phi}$$

$$I = \int_0^{2\pi} \int_{\pi/3}^{\pi/2} \int_0^2 \frac{r^2 \sin \phi dr d\phi d\theta}{r^2} +$$

$$\int_0^{2\pi} \int_{\pi/2}^{\pi/3} \int_{\frac{3 \cos \phi}{\sin^2 \phi}}^2 \frac{r^2 \sin \phi dr d\phi d\theta}{r^2} = \underline{\underline{2\pi - 6 \ln\left(\frac{\sqrt{3}}{2}\right) \pi}}$$

9) Evaluate the volume of the solid that is enclosed by $z = \sqrt{3x^2 + 3y^2}$, $z = 1$ and $z = 2$ by using spherical coordinates.

$$z = \sqrt{3x^2 + 3y^2} \Rightarrow z^2 = 3x^2 + 3y^2 \text{ cone. } (z \geq 0)$$



$$V = \iiint dV$$

with spherical coord.

$$V = \iiint r^2 \sin \phi dr d\phi d\theta$$

$$V = \int_0^{2\pi} \int_0^{\pi/6} \int_{1/\cos \phi}^{2/\cos \phi} r^2 \sin \phi dr d\phi d\theta$$

$$V = \int_0^{2\pi} \int_0^{\pi/6} \left[\frac{r^3}{3} \right]_{1/\cos \phi}^{2/\cos \phi} \sin \phi d\phi d\theta$$

$$V = \frac{1}{3} \int_0^{2\pi} \int_0^{\pi/6} \left[\frac{8}{\cos^3 \phi} - \frac{1}{\cos^3 \phi} \right] \sin \phi d\phi d\theta$$

$$V = \frac{7}{3} \int_0^{2\pi} \int_0^{\pi/6} \frac{1}{\cos^3 \phi} \sin \phi d\phi d\theta = \frac{7}{3} \int_0^{2\pi} \left[-\frac{1}{2 \cos^2 \phi} \right]_0^{\pi/6} d\theta$$

$$V = \frac{7}{6} \int_0^{2\pi} \left(\frac{4}{3} - 1 \right) d\theta = \frac{7}{6} \cdot \frac{1}{3} \cdot 2\pi = \frac{7\pi}{9}$$

10) Determine whether the integral $I = \int_0^4 \frac{dx}{\sqrt[3]{4x^2-x^3}}$ and evaluate if it's convergent.

$I = \int_0^4 \frac{dx}{\sqrt[3]{4x^2-x^3}}$, function $\frac{1}{\sqrt[3]{4x^2-x^3}}$ is not cont. at $x=0$ and $x=4$.

$$I = \underbrace{\int_0^2 \frac{dx}{\sqrt[3]{4x^2-x^3}}}_{I_1} + \underbrace{\int_2^4 \frac{dx}{\sqrt[3]{4x^2-x^3}}}_{I_2}$$

for I_1 : $I_1 = \int_0^2 \frac{dx}{\sqrt[3]{4x^2-x^3}}$ 2nd type, choose

$\int_0^2 \frac{dx}{\sqrt{x^2}}$ p integral with $p = \frac{2}{3} < 1$ convergent.

$$\lim_{x \rightarrow 0^+} \frac{f(x)}{g(x)} = \lim_{x \rightarrow 0^+} \frac{\sqrt[3]{x^2}}{\sqrt[3]{4x^2-x^3}} = \lim_{x \rightarrow 0^+} \frac{\sqrt[3]{x^2}}{\sqrt[3]{x^2} \cdot \sqrt[3]{4-x}} = \frac{1}{\sqrt[3]{4}} \neq 0$$

and since $\int_0^2 \frac{dx}{\sqrt{x^2}}$ converges then I_1 is convergent.

for I_2 : $I_2 = \int_2^4 \frac{dx}{\sqrt[3]{4x^2-x^3}}$, 2nd type, choose

$\int_2^4 \frac{dx}{\sqrt[3]{4-x}}$ p-integral with $p = 1/3 < 1$ convergent.

$$\lim_{x \rightarrow 4^-} \frac{f(x)}{g(x)} = \lim_{x \rightarrow 4^-} \frac{\sqrt[3]{4-x}}{\sqrt[3]{4x^2-x^3}} = \lim_{x \rightarrow 4^-} \frac{\sqrt[3]{4-x}}{\sqrt[3]{x^2} \cdot \sqrt[3]{4-x}} = \frac{1}{\sqrt[3]{24}} \neq 0$$

and since $\int_2^4 \frac{dx}{\sqrt[3]{4-x}}$ converges then I_2 is convergent.

Since I_1 and I_2 are convergent then $I = I_1 + I_2$ is convergent.

$$\int_0^4 \frac{dx}{\sqrt[3]{4x^2 - x^3}} = \int_0^4 \frac{dx}{\sqrt[3]{x^2} \cdot \sqrt[3]{4-x}} = \int_0^4 \frac{dx}{\sqrt[3]{x^2} \cdot \sqrt[3]{4} \cdot \sqrt[3]{1-x/4}}$$

let $\frac{x}{4} = t \Rightarrow dx = 4dt$

$$I = \int_0^1 \frac{4dt}{\sqrt[3]{16t^2} \cdot \sqrt[3]{4} \cdot \sqrt[3]{1-t}} = \frac{4}{\sqrt[3]{4}} \cdot \frac{1}{\sqrt[3]{16}} \int_0^1 t^{-2/3} \cdot (1-t)^{-1/3} dt$$

$$I = \frac{4}{\sqrt[3]{4^3}} \int_0^1 \underbrace{t^{-2/3} \cdot (1-t)^{-1/3} dt}_{\text{beta integral}}, \quad \begin{array}{l} m-1 = -2/3 \\ m = 1/3 \\ n-1 = -1/3 \\ n = 2/3 \end{array}$$

$$I = \beta\left(\frac{1}{3}, \frac{2}{3}\right) = \frac{\Gamma(1/3) \cdot \Gamma(2/3)}{\Gamma\left(\frac{1}{3} + \frac{2}{3}\right)} = \frac{\Gamma\left(\frac{1}{3}\right) \cdot \Gamma\left(\frac{2}{3}\right)}{1}$$

ii) Is $I = \int_0^\infty \frac{x dx}{x^4 + 4}$ convergent? If yes, evaluate it, by using gamma-beta functions.

$I = \int_0^\infty \frac{x dx}{x^4 + 4}$ improper intg of 1st type. Before applying p-integral, we need to use;

$$I = \int_0^\infty \frac{x dx}{x^4 + 4} = \underbrace{\int_0^1 \frac{x dx}{x^4 + 4}}_{\text{proper intg. (convergent)}} + \int_1^\infty \frac{x dx}{x^4 + 4}$$

For $\int_1^{\infty} \frac{x dx}{x^4 + 4}$ let choose $\int_1^{\infty} \frac{dx}{x^3}$ p-integral

with $p=3 > 1$ convergent,

$$\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = \lim_{x \rightarrow \infty} \frac{x \cdot x^3}{x^4 + 4} = 1 \neq 0 \text{ and since}$$

$$\int_1^{\infty} \frac{dx}{x^3} \text{ converges then } \int_1^{\infty} \frac{x dx}{x^4 + 4} \text{ also converges.}$$

Followingly $I = \int_1^{\infty} \frac{x dx}{x^4 + 4}$ is convergent. Then we can evaluate by applying gamma-beta functions.

$$I = \int_0^{\infty} \frac{x dx}{x^4 + 4} = \frac{1}{4} \int_0^{\infty} \frac{x dx}{1 + \frac{x^4}{4}} = \frac{1}{4} \int_0^{\infty} \frac{x dx}{1 + \left(\frac{x^2}{2}\right)^2}$$

$$\frac{x^2}{2} = t \Rightarrow x^2 = 2t \Rightarrow 2x dx = 2dt$$

$$I = \frac{1}{4} \int_0^{\infty} \frac{dt}{1+t^2}, \text{ now use } \begin{cases} \frac{1}{1+t^2} = y \Rightarrow 1+t^2 = \frac{1}{y} \\ \frac{-2t dt}{(1+t^2)^2} = dy \end{cases} \quad t = \sqrt{\frac{1}{y} - 1}$$

$$I = \frac{1}{4} \int_1^0 y \cdot \frac{(1+t^2)^2}{-2t} dy = \frac{1}{8} \int_0^1 y \cdot \frac{1}{y^2} \cdot \frac{dy}{\sqrt{1-y}}$$

$$I = \frac{1}{8} \int_0^1 \frac{1}{y} \cdot \sqrt{y} \cdot \frac{dy}{\sqrt{1-y}} = \frac{1}{8} \int_0^1 \underbrace{y^{-1/2} \cdot (1-y)^{-1/2}}_{\text{beta function}} dy$$

$$m-1 = -\frac{1}{2} \Rightarrow m = \frac{1}{2}, \quad n-1 = -\frac{1}{2} \Rightarrow n = \frac{1}{2}$$

$$I = \frac{1}{8} \beta\left(\frac{1}{2}, \frac{1}{2}\right) = \frac{1}{8} \cdot \frac{\Gamma\left(\frac{1}{2}\right) \cdot \Gamma\left(\frac{1}{2}\right)}{\Gamma(1)} = \frac{1}{8} \sqrt{\pi} \cdot \sqrt{\pi}$$

$$I = \frac{\pi}{8}$$

12) Determine whether the integral $I = \int_1^{\infty} \frac{dx}{\ln^2 x}$ is convergent or not. If yes, evaluate it by using gamma and beta functions.

$$I = \int_1^{\infty} \frac{dx}{\ln^2 x} \quad \text{mixed type since } \frac{1}{\ln^2 x} \text{ is discont. at } x=1.$$

$$I = \underbrace{\int_1^2 \frac{dx}{\ln^2 x}}_{I_1} + \underbrace{\int_2^{\infty} \frac{dx}{\ln^2 x}}_{I_2}$$

for I_1 : $I_1 = \int_1^2 \frac{dx}{\ln^2 x}$ 2nd type. let choose

$\int_1^2 \frac{dx}{x-1}$ p-integral with $p=1$, diverges.

$$\lim_{x \rightarrow 1^+} \frac{f(x)}{g(x)} = \lim_{x \rightarrow 1^+} \frac{\frac{0}{0}}{\frac{1}{\ln^2 x}} \stackrel{0}{=} \lim_{x \rightarrow 1^+} \frac{1}{\frac{2 \ln x}{x}} = \lim_{x \rightarrow 1^+} \frac{x}{2 \ln x} = \infty$$

and since $\int_1^2 \frac{dx}{x-1}$ diverges then I_1 diverges.

Then directly we can conclude that I is divergent and so we can not use gamma or beta functions.

13) Evaluate $I = \int_0^{\infty} \frac{dx}{(x^2+1)^3}$ by using the Integral $\int_0^{\infty} \frac{dx}{x^2+a}$.

$$\int_0^{\infty} \frac{dx}{x^2+a} = \lim_{k \rightarrow \infty} \int_0^k \frac{dx}{x^2+a} = \lim_{k \rightarrow \infty} \frac{1}{\sqrt{a}} \arctan \frac{x}{\sqrt{a}} \Big|_0^k$$

$$= \lim_{k \rightarrow \infty} \frac{1}{\sqrt{a}} \cdot \underbrace{\arctan \frac{k}{\sqrt{a}}}_{\pi/2} - 0$$

$$= \frac{\pi}{2\sqrt{a}} \Rightarrow \int_0^{\infty} \frac{dx}{x^2+a} = \frac{\pi}{2\sqrt{a}} = \frac{\pi}{2} \cdot a^{-1/2}$$

Now use the Leibnitz's Rule;

$$\int_0^{\infty} \frac{-1}{(x^2+a)^2} dx = \frac{\pi}{2} \cdot \left(-\frac{1}{2}\right) a^{-3/2}$$

derivative wrt a : parameter

$$\int_0^{\infty} \frac{2}{(x^2+a)^3} dx = \frac{\pi}{2} \left(-\frac{1}{2}\right) \left(-\frac{3}{2}\right) a^{-5/2}$$

for $a=1$

$$\int_0^{\infty} \frac{2}{(x^2+1)^3} dx = \frac{\pi}{2} \cdot \frac{3}{4} \cdot 1 = \frac{3\pi}{8}$$

$$I = \int_0^{\infty} \frac{dx}{(x^2+1)^3} = \frac{3\pi}{16} //$$